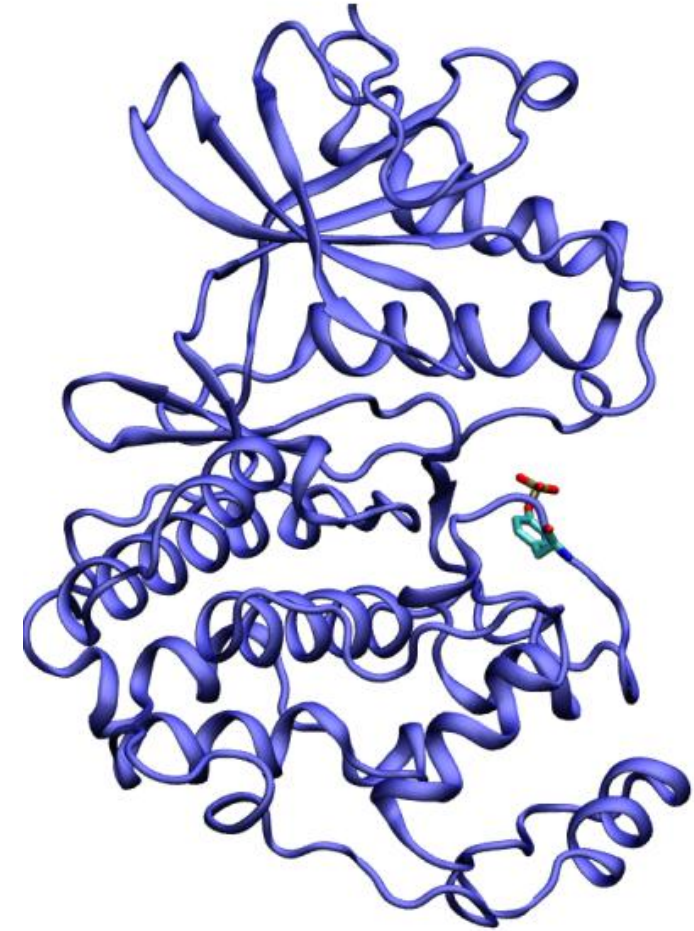
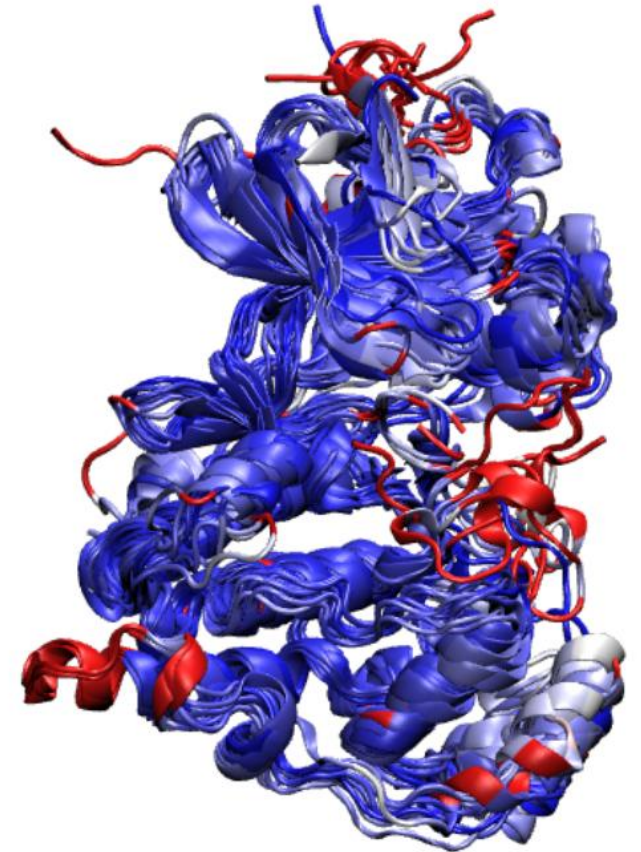
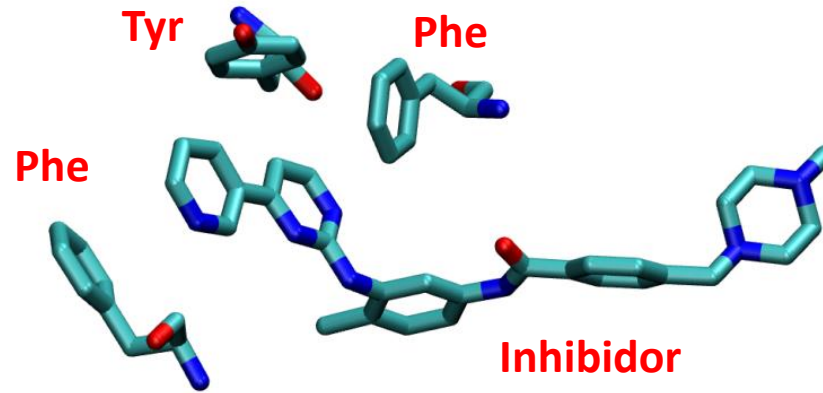


TP

Análisis de proteínas y alineamiento estructural con VMD

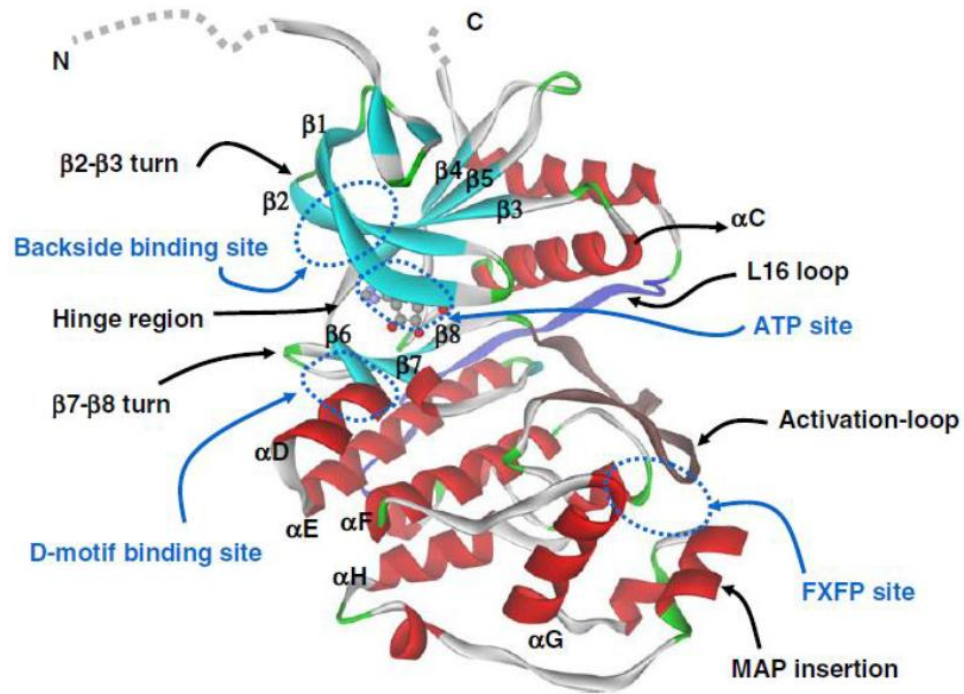


Representación proteína

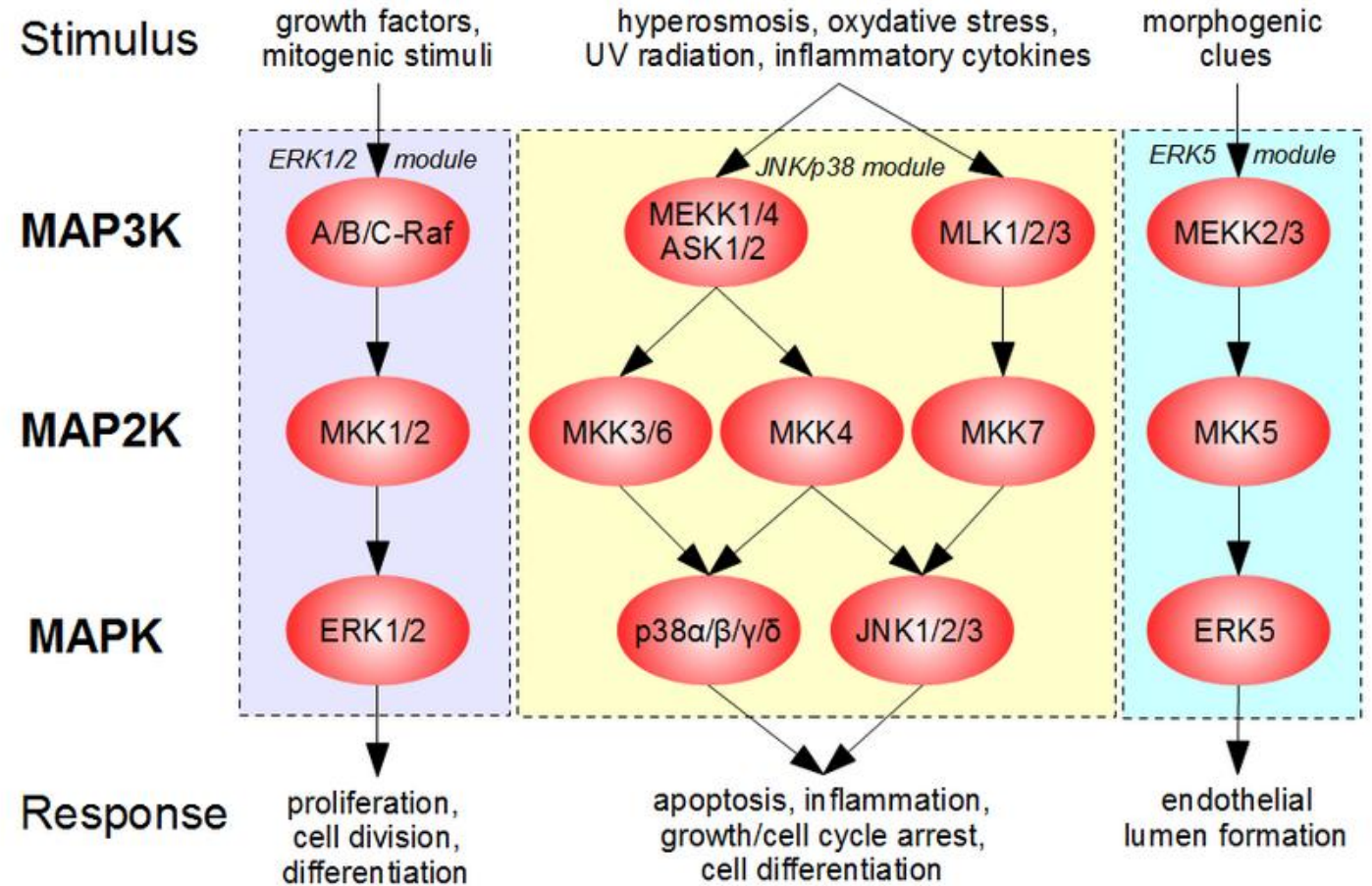


Alineamiento estructural de proteínas

MAPK QUINASAS



Simplified overview of mammalian MAPK cascades



PDB ID: código de 4 letras/números

RCSB

PDB

PROTEIN DATA BANK

219,869 Structures from the PDB

1,068,577 Computed Structure Models (CSM)

3D Structures

Enter search term(s), Entry ID(s), or sequence

Include CSM

Q

Advanced Search

Browse Annotations

Help

PDB-101

PDB

EMDataResource

NAKB

wwPDB Foundation

PDB-Dev

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Welcome

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Search

Visualize

Analyze

Download

Learn

RCSB Protein Data Bank (RCSB PDB) enables breakthroughs in science and education by providing access and tools for exploration, visualization, and analysis of:

Experimentally-determined 3D structures from the **Protein Data Bank (PDB)** archive

Computed Structure Models (CSM) from AlphaFold DB and ModelArchive

These data can be explored in context of external annotations providing a structural view of biology.

Explore NEW Features

PDB

PROTEIN DATA BANK

NEW!

Q

SEARCH

🔬

LABORATORY

📄

DOCUMENT

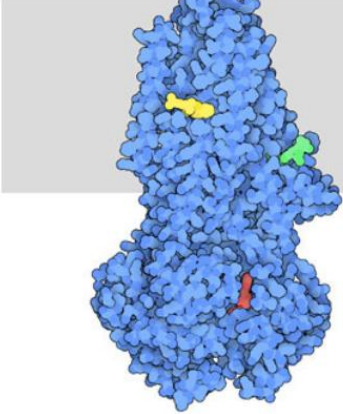
💡

IDEA

PDB-101

Training Resources

May Molecule of the Month



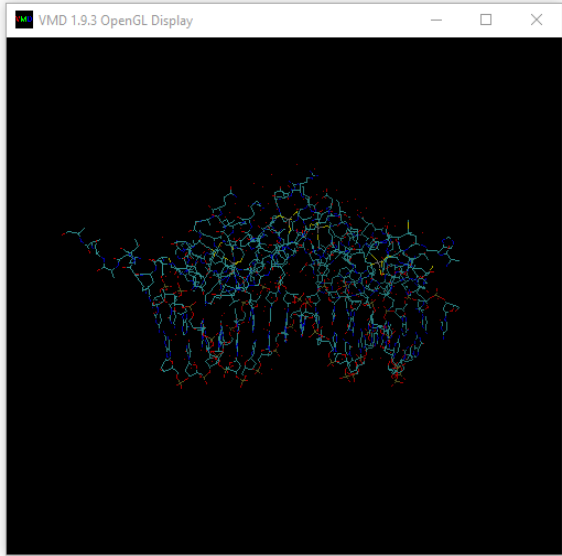
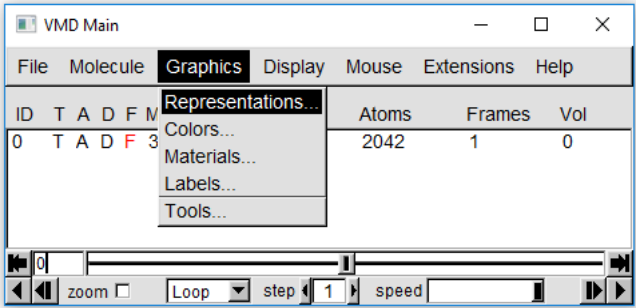
CFTR and Cystic Fibrosis

BLAST contra PDB
UNIPROT

HEADER	TRANSFERASE	28-MAR-08	3COI
TITLE	CRYSTAL STRUCTURE OF P38DELTA KINASE		
COMPND	MOL_ID: 1;		
COMPND	2 MOLECULE: MITOGEN-ACTIVATED PROTEIN KINASE 13;		
COMPND	3 CHAIN: A;		
COMPND	4 SYNONYM: STRESS-ACTIVATED PROTEIN KINASE 4, MITOGEN-		
COMPND	5 ACTIVATED PROTEIN KINASE P38 DELTA, MAP KINASE P38 DELTA;		
COMPND	6 EC: 2.7.11.24;		
COMPND	7 ENGINEERED: YES		
SOURCE	MOL_ID: 1;		
SOURCE	2 ORGANISM_SCIENTIFIC: HOMO SAPIENS;		
SOURCE	3 ORGANISM_COMMON: HUMAN;		
SOURCE	4 ORGANISM_TAXID: 9606;		
SOURCE	5 STRAIN: HEPG2;		
SOURCE	6 GENE: MAPK13, PRKM13, SAPK4;		
SOURCE	7 EXPRESSION_SYSTEM: ESCHERICHIA COLI BL21(DE3);		
SOURCE	8 EXPRESSION_SYSTEM_TAXID: 469008;		
SOURCE	9 EXPRESSION_SYSTEM_STRAIN: BL21(DE3);		

Formato PDB

	# átomo	nombre átomo	nombre residuo	cadena	# residuo	coordenadas xyz			B-factor	tipo de átomo	
proteína	ATOM	1	N	ALA	A 523	65.336	11.629	-15.337	1.00	126.58	N
	ATOM	2	CA	ALA	A 523	64.511	11.427	-16.562	1.00	127.19	C
	ATOM	3	C	ALA	A 523	63.097	11.972	-16.357	1.00	127.59	C
	ATOM	4	O	ALA	A 523	62.805	12.597	-15.339	1.00	127.91	O
	ATOM	5	CB	ALA	A 523	64.460	9.946	-16.917	1.00	126.92	C
	ATOM	6	N	THR	A 524	62.231	11.723	-17.333	1.00	127.65	N
	ATOM	7	CA	THR	A 524	60.847	12.186	-17.275	1.00	126.90	C
	ATOM	8	C	THR	A 524	60.148	11.848	-15.965	1.00	125.83	C
	ATOM	9	O	THR	A 524	60.262	10.743	-15.448	1.00	125.82	O
	ATOM	10	CB	THR	A 524	60.010	11.599	-18.432	1.00	127.04	C
	ATOM	11	OG1	THR	A 524	60.456	12.145	-19.672	1.00	127.57	O
	ATOM	12	CG2	THR	A 524	58.528	11.917	-18.248	1.00	126.40	C
	ATOM	13	N	LEU	A 525	59.426	12.836	-15.451	1.00	124.50	N
	ATOM	14	CA	LEU	A 525	58.651	12.692	-14.230	1.00	123.18	C
	ATOM	15	C	LEU	A 525	57.674	13.860	-14.209	1.00	122.11	C
	ATOM	16	O	LEU	A 525	57.909	14.885	-14.847	1.00	122.88	O
	ATOM	17	CB	LEU	A 525	59.551	12.724	-12.984	1.00	122.43	C
	ATOM	18	CG	LEU	A 525	59.513	11.422	-12.141	1.00	122.01	C
	ATOM	19	CD1	LEU	A 525	60.563	10.447	-12.647	1.00	121.38	C
	ATOM	20	CD2	LEU	A 525	59.762	11.742	-10.678	1.00	122.06	C
	ATOM	21	N	PRO	A 526	56.570	13.728	-13.464	1.00	120.65	N
	ATOM	22	CA	PRO	A 526	56.146	12.598	-12.637	1.00	119.13	C
	ATOM	23	C	PRO	A 526	56.247	11.133	-13.096	1.00	116.64	C
	ATOM	24	O	PRO	A 526	56.260	10.790	-14.275	1.00	117.64	O
	ATOM	25	CB	PRO	A 526	54.736	12.971	-12.276	1.00	120.36	C
	ATOM	26	CG	PRO	A 526	54.857	14.435	-12.050	1.00	120.80	C
	ATOM	27	CD	PRO	A 526	55.668	14.873	-13.256	1.00	120.95	C
TER											
ligando agua etc	HETATM	4579	O2	DEX	401	16.637	25.377	-12.444	1.00	58.44	O
	HETATM	4580	O3	DEX	401	19.199	22.632	-15.880	1.00	53.96	O
	HETATM	4581	O4	DEX	401	16.766	23.931	-17.924	1.00	59.24	O
	HETATM	4582	O5	DEX	401	15.148	22.063	-17.018	1.00	59.47	O
	HETATM	4583	O	HOH	1	21.275	37.841	-16.997	1.00	59.66	O
	HETATM	4584	O	HOH	2	42.952	4.257	17.028	1.00	56.68	O
	HETATM	4585	O	HOH	3	30.648	30.768	-21.751	1.00	64.70	O
	HETATM	4586	O	HOH	4	41.876	4.799	4.926	1.00	55.12	O
	HETATM	4587	O	HOH	5	37.987	37.282	-16.691	1.00	73.89	O
	HETATM	4588	O	HOH	6	39.981	2.750	16.073	1.00	49.77	O
	HETATM	4589	O	HOH	7	25.696	52.416	-14.906	1.00	64.00	O
	HETATM	4590	O	HOH	8	21.045	4.893	-5.604	1.00	57.17	O
	HETATM	4591	O	HOH	9	8.549	27.690	-1.842	1.00	67.90	O
END											

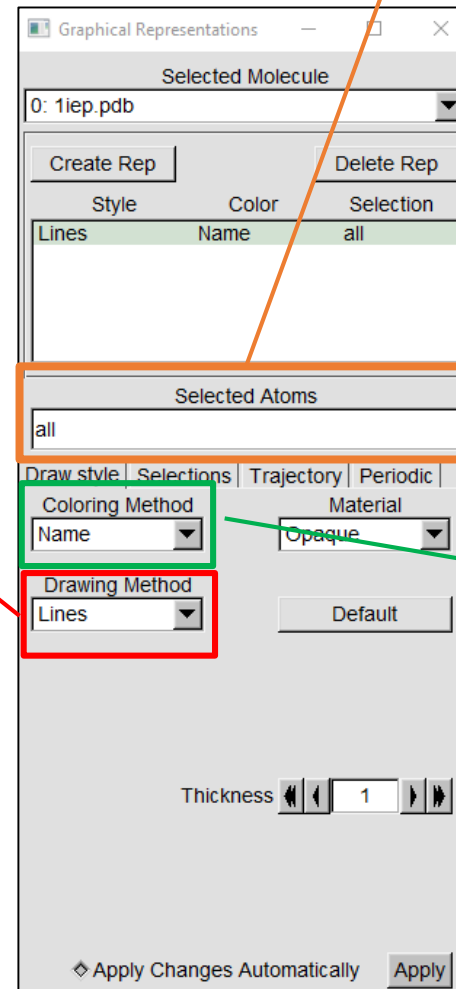


VMD: Visual Molecular Dynamics

# átomo	nombre átomo	nombre residuo	cadena	# residuo	coordenadas xyz
1	N	ALA	A	523	65.336 11.629 -15.337
2	CA	ALA	A	523	64.511 11.427 -16.562
3	C	ALA	A	523	63.097 11.972 -16.357
4	O	ALA	A	523	62.805 12.597 -15.339
5	CB	ALA	A	523	64.460 9.946 -16.917
6	N	THR	A	524	62.231 11.723 -17.333

- # átomo = *index*
- nombre de átomo = *name*
- nombre residuo = *resname*
- cadena = *chain*
- # residuo = *resid*

- Drawing Method
- Lines
 - Bonds
 - DynamicBonds
 - HBonds
 - Points
 - VDW
 - CPK
 - Licorice
 - Polyhedra
 - Trace
 - Tube
 - Ribbons
 - NewRibbons
 - Cartoon
 - NewCartoon



- Coloring Method
- Name
 - Type
 - Element
 - ResName
 - ResType
 - ResID
 - Chain
 - SegName
 - Conformation
 - Molecule
 - Secondary Structure
 - ColorID
 - Beta
 - Occupancy
 - Mass
 - Charge

Crea una nueva
representación

Graphical Representations

Selected Molecule

0: 3G6P.pdb

Create Rep Delete Rep

Style	Color	Selection
NewCartoon	Name	protein
Licorice	Name	resid 474

Selected Atoms

resid 474

Draw style | Selections | Trajectory | Periodic

Coloring Method

Name

Material

Opaque

Drawing Method

Licorice

Default

Sphere Resolution 12

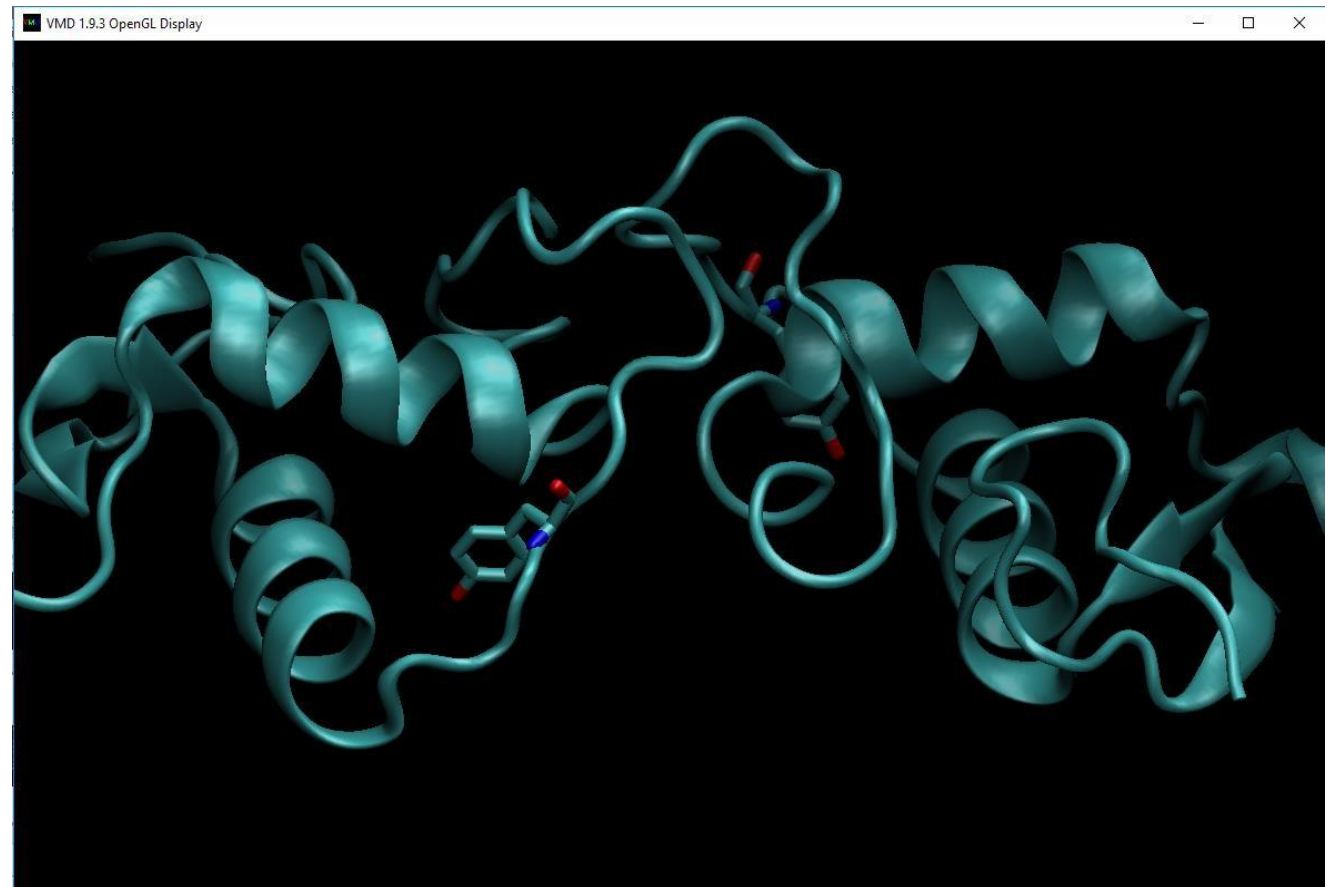
Bond Radius 0.3

Bond Resolution 12

◇ Apply Changes Automatically

Apply

Rep.1
Rep. 2



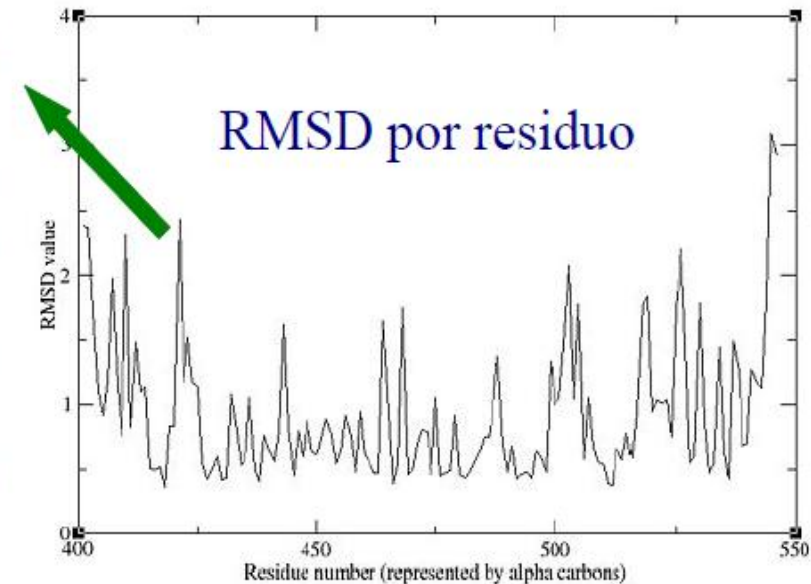
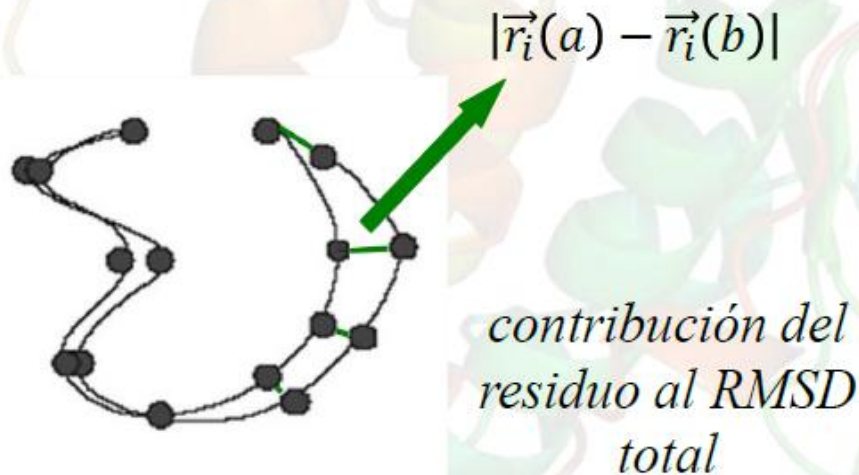
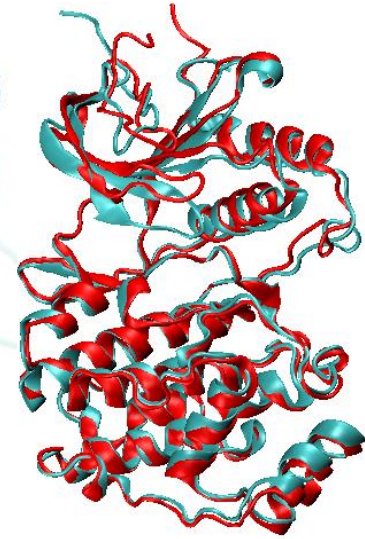
Parámetros de similitud estructural RMSD

RMSD = distancia promedio entre los átomos de las estructuras alineadas.

$$RMSD = \sqrt{\frac{1}{N} \sum_{i=1}^N |\vec{r}_i(a) - \vec{r}_i(b)|^2}$$

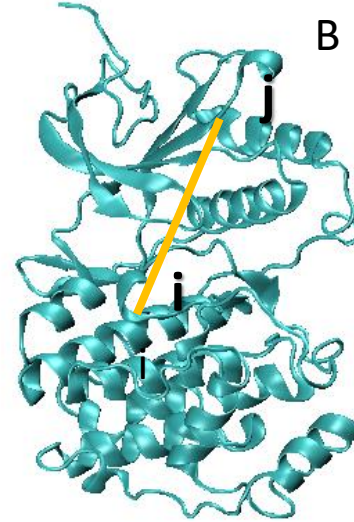
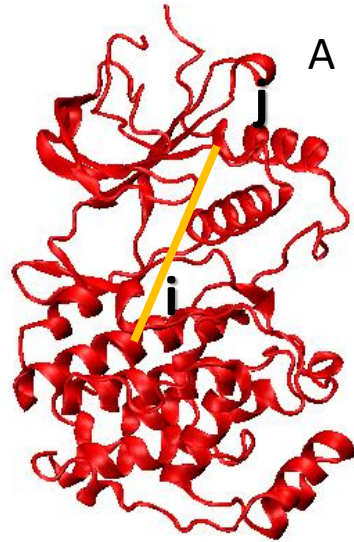
se busca **minimizar**

N = número de átomos / a = proteína de referencia / b = proteína alineada



Parámetros de similitud de dos estructuras Q

Correlación de distancias
internas entre residuos
equivalentes en dos
estructuras



No requiere
superposición
estructural

$$Q = \frac{2}{(N-1)(N-2)} \sum_{i < j-1}^{N-1} \exp \left[-\frac{(r_{ij}^A - r_{ij}^B)^2}{2s_{ij}^2} \right]$$



se busca **maximizar**



N = número de residuos en proteína de referencia / r_{ij} = distancia entre el C α de un residuo y otro en la proteína alineada (la suma evalúa las distancias al resto de los C α -todos contra todos excepto los contiguos-) / $r_{ij,nat}$ = distancia entre el C α de un residuo y otro en la proteína de referencia / s_{ij} = desviación estándar de la distribución de r_{ij} .

(estructuras no alineadas) $0 < Q \leq 1$ (estructuras idénticas)

Superposición de estructuras

Alineamiento de
secuencia

Residuos equivalentes
conocidos

```

105 S I R I G S K G D V F V I R E P F I S C S H L E C R T F F L T G A L L N D K H S N G T V K D R S P Y R 156
23 S I R I G S K G D V F V I R E P F I S C S H L E C R T F F L T G A L L N D K H S N G T V K D R S P H R 74

157 T L M S C P L G E A P S P Y N S R F E S V A W S A S A C H D G M S W L T I G I S G P D N G A V A V L K Y 208
75 T L M S C P V G E A P S P Y N S R F E S V A W S A S A C H D G T S W L T I G I S G P D N G A V A V L K Y 126

209 N G I I T E T I K S W K K R I L R T Q E S E C V C M N G S C F T I M T D G P S N G A A S Y K I F K I E K 260
127 N G I I T D T I K S W R N N I L R T Q E S E C A C V N G S C F T V M T D G P S N G Q A S Y K I F K M E K 178

261 G K V T K T I E L N A P N F H Y E E C S C Y P D T G T V M C V C R D N W H G S N R P W V S F N Q N L D Y 312
179 G K V V K S V E L D A P N Y H Y E E C S C Y R N A G E I T C V C R D N W H G S N R P W V S F N Q N L E Y 230

313 D I G Y I C S G V F G D N P R P K D G E S C N P V T V D G A D G V K G F S Y K Y G N G V W I G R T K S 364
231 D I G Y I C S G V F G D N P R P N D G T S C G P V S N G A Y G V K G F S F K Y G N G V W I G R T K S 282

365 N R L K K F E M I W D P N G W T N T D S F S V K Q D V V A I T D W S G Y S G S F V Q H P E L T G L D 416
283 T N S S G F E M I W D P N G W T E T D S F S V K Q D I V A I T D W S G Y S G S F V Q H P E L T G L D 334

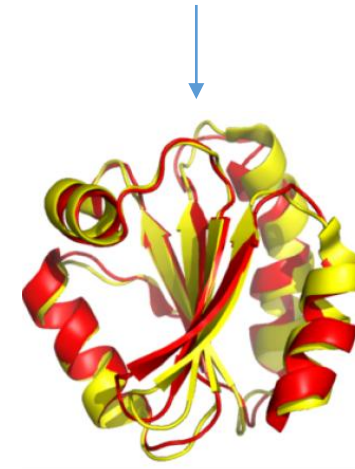
417 C I R P C F W V E L V R G L P R E N T T I W T S G S S I S F C G V N S G T A N W S . . . . . 457
335 C I R P C F W V E L I R G R P K E - S T I W T S G S S I S F C G V N S D T V G W S W F D A E L F T I 385
    
```

Minimización de la
distancia y RMSD entre
residuos equivalentes



Alineamiento estructural

Superposición óptima en base
a puntaje por similitud de
estructura local y
programación dinámica



Calculo de
RMSD

Definición de residuos
equivalentes por cercanía
espacial

Alineamiento de
secuencia derivado

```

105 S I R I G S K G D V F V I R E P F I S C S H L E C R T F F L T G A L L N D K H S N G T V K D R S P Y R 156
23 S I R I G S K G D V F V I R E P F I S C S H L E C R T F F L T G A L L N D K H S N G T V K D R S P H R 74

157 T L M S C P L G E A P S P Y N S R F E S V A W S A S A C H D G M S W L T I G I S G P D N G A V A V L K Y 208
75 T L M S C P V G E A P S P Y N S R F E S V A W S A S A C H D G T S W L T I G I S G P D N G A V A V L K Y 126

209 N G I I T E T I K S W K K R I L R T Q E S E C V C M N G S C F T I M T D G P S N G A A S Y K I F K I E K 260
127 N G I I T D T I K S W R N N I L R T Q E S E C A C V N G S C F T V M T D G P S N G Q A S Y K I F K M E K 178

261 G K V T K T I E L N A P N F H Y E E C S C Y P D T G T V M C V C R D N W H G S N R P W V S F N Q N L D Y 312
179 G K V V K S V E L D A P N Y H Y E E C S C Y R N A G E I T C V C R D N W H G S N R P W V S F N Q N L E Y 230

313 D I G Y I C S G V F G D N P R P K D G E S C N P V T V D G A D G V K G F S Y K Y G N G V W I G R T K S 364
231 D I G Y I C S G V F G D N P R P N D G T S C G P V S N G A Y G V K G F S F K Y G N G V W I G R T K S 282

365 N R L K K F E M I W D P N G W T N T D S F S V K Q D V V A I T D W S G Y S G S F V Q H P E L T G L D 416
283 T N S S G F E M I W D P N G W T E T D S F S V K Q D I V A I T D W S G Y S G S F V Q H P E L T G L D 334

417 C I R P C F W V E L V R G L P R E N T T I W T S G S S I S F C G V N S G T A N W S . . . . . 457
335 C I R P C F W V E L I R G R P K E - S T I W T S G S S I S F C G V N S D T V G W S W F D A E L F T I 385
    
```

STAMP (STructural Alignment of Multiple Proteins)

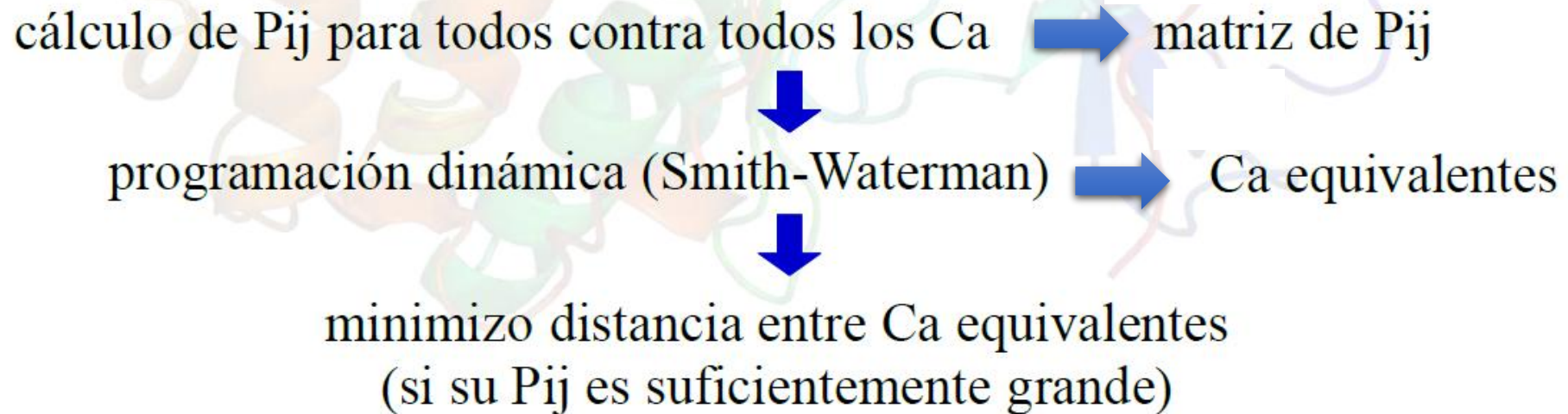
$$P_{ij} = \exp\left(-\frac{d_{ij}^2}{C_1}\right) \exp\left(-\frac{s_{ij}^2}{C_2}\right)$$

P_{ij} = probabilidad de que un residuo i en una proteína sea equivalente a un residuo j en otra proteína

d_{ij} = distancia entre los Ca

s_{ij} = medida de la diferencia de conformación local del backbone para los residuos i y j

C = constantes (pesos para distancia / conformación)



Alineamiento estructural vs. alineamiento de secuencias

Árboles filogenéticos

